

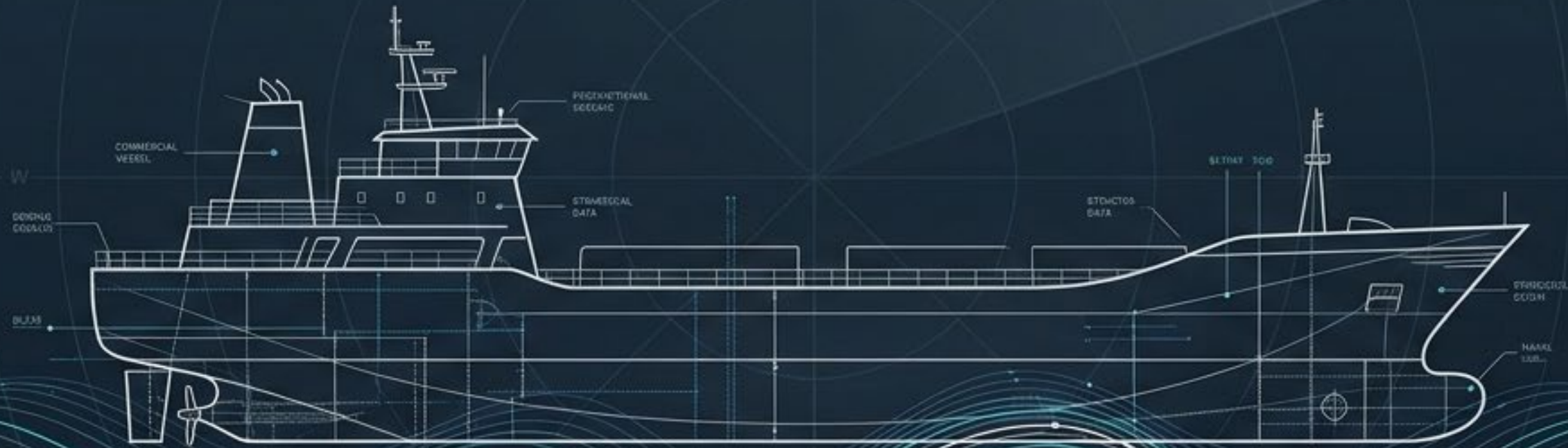


The Maritime Decarbonization Command Center

A unified platform to calculate, verify, and optimize compliance across IMO CII, EEXI, and EU MRV/FuelEU frameworks.

NavCIIfuel: The Ultimate Maritime Compliance Suite

Navigating the future of vessel performance.



Seamless Access to a Unified Compliance Toolkit

A unified portal providing fast, regulation-aligned calculations ahead of reporting deadlines.

1. The Gateway



MARITIME CALCULATOR

 Continue with Google

 Continue as Guest

By continuing you agree to our Terms & Privacy Policy.
Guest access uses sample vessel demo data.



2. The Command Center



NAV

Go to NavStab →

Naval Architecture & Vessel
Performance

Ship stability calculations,
hydrostatics & loading conditions



CII

EEXI

Go to CII & EEXI Calculator →

Carbon Intensity & Energy
Efficiency Index

IMO CII rating, EEXI compliance &
decarbonization tracking



FUEL

Go to FuelEU Calculator →

Fuel Efficiency, FuelEU Maritime
& Emissions Reduction

GHG intensity compliance, penalty &
pooling under EU Regulation

The Compliance Command Center

One unified platform to master naval architecture, carbon intensity, and regional maritime regulations.



NAV

Go to NavStab →

Naval Architecture & Vessel Performance
Ship stability calculations, hydrostatics & loading conditions.



CII

EEXI

Go to CII & EEXI Calculator →

Carbon Intensity & Energy Efficiency Index
IMO CII rating, EEXI compliance & decarbonization tracking.



FUEL

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Fuel Efficiency, FuelEU Maritime & Emissions Reduction
GHG intensity compliance, penalty & pooling under EU Regulation.

NavStab: The Physical Foundation

Before compliance, there is physics. Precision naval engineering at the hull level.

Stability Calculations

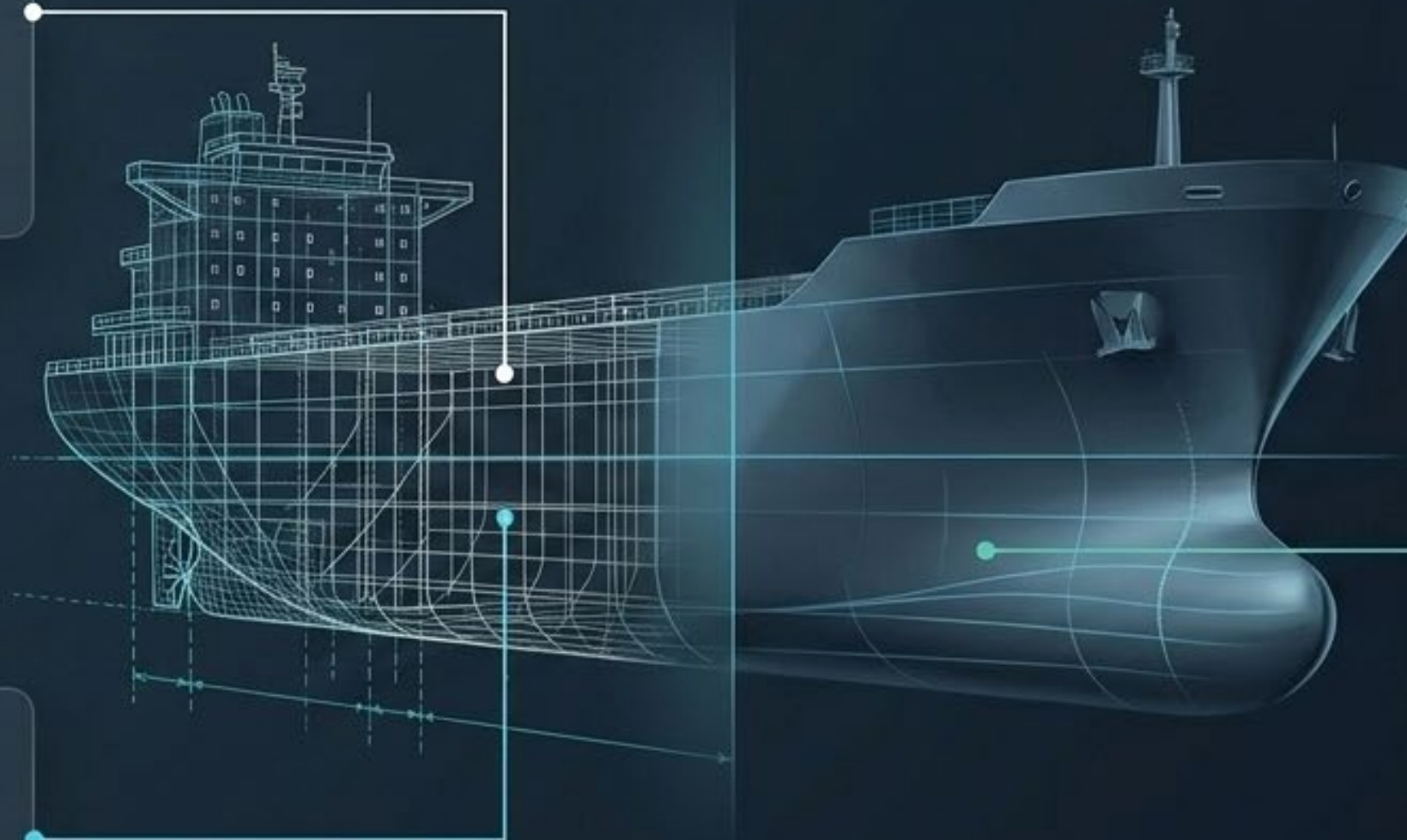
Real-time **center of gravity** and intact/damage stability modeling.

Hydrostatics

Precision volume and buoyancy metrics.

Loading Conditions

Dynamic cargo and weight distribution stress testing.



The NavStab Architecture Lifecycle

Pillar 1: Hull Geometry

The Foundation



Design Coefficients



CB CW CP CN DISPLACEMENT WPA TPC



TPC & Displacement



TPC FWA DWA SALT / FRESH VOLUME

Pillar 2: Intact Stability

Operational Planning



Transverse Stability (GM, Loll, Inclining)



GM KM KB BM LOLL ANGLE INCLINING



Trim & Drafts



MCTC LCF BML CHANGE OF TRIM NEW DRAFTS

Pillar 3: Special Dynamics

Edge Conditions



Free Surface Effect (FSE)



FSM LOSS OF GM TANK DIVISIONS EFFECTIVE GM



Ship Squat



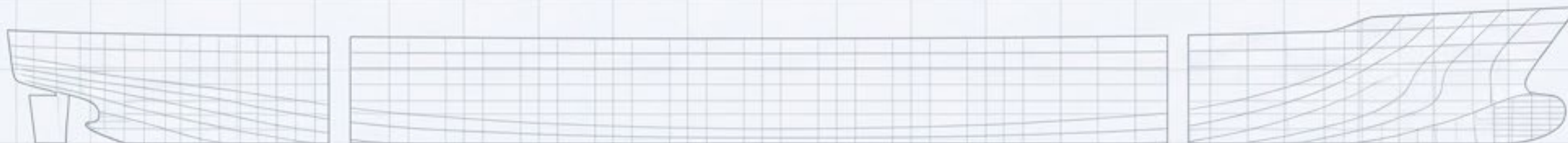
BARRAS FORMULA SPEED VS SQUAT TABLE UKC SAFETY CHECK



Drydocking



UPTHURST P CRITICAL GM RIGHTING MOMENT



The Efficiency Paradigm: EEXI vs. CII

Understanding the distinction between your technical baseline and your operational reality.

INDEX 01 / DESIGN

KWH · GCO₂



EEXI

Energy Efficiency Existing Ship Index —
one-time technical rating of design efficiency.

INDEX 02 / OPERATIONAL

AER/CGDIST · GCO₂/T·NM



CII

Carbon Intensity Indicator — annual
operational rating, A through E.

EEXI Toolkit: Securing the Baseline

Verify attainment and ensure accurate decarbonization compliance.

Control Panel



EEXI Maritime Calculator

EEXI Maritime Compliance Toolkit

Choose calculation module

EEXI Maritime Calculator

Open the EEXI Maritime Calculator to check the EEXI compliance and report generation.



SFC ISO Correction Tool

Open the SFC ISO Correction Tool to check the SFC for both 2-stroke and 4-stroke engine types.



Check the Cf of the biofuel

Open the biofuel conversion factor reference page for fuel-specific Cf checking.



CII Portal: Operational Excellence Across Jurisdictions

Track, verify, and defend your A-to-E rating on a global scale.

CII Maritime Calculator



CII Maritime Compliance Toolkit



Choose calculation module

1

CII Maritime Calculator

Open the CII calculator to review vessel carbon intensity performance and rating output.



2

Canadian CII Maritime Calculator

Open the Canadian CII calculator for Transport Canada compliant vessel carbon intensity ratings.



3

IMO DCS & CII Verification Tool

Open the IMO DCS & CII Verification Tool for vessel data collection and CII checks.



Disparate data funnels into regulation-ready exports

Output Engine



Standardized DCS/FoC Annual Report

Name of the ship		Gas carrier - N/A	
IMO Number		N/A	
EEDI/EEEXI		Attained EEDI/EEEXI	
Fuel oil consumed per consumer type			
Type		Quantity	Monitoring Method
Main Engine	HFO	36899.000t	--
	LNG	3450.000t	
Aux Engine	HFO	5000.000t	--
	LNG	434130.000t	
Inert Gas	HFO	3455000.000t	--
Boiler	HFO	6000.000t	--
	LNG	5000.000t	

Standardized DCS/FoC Annual Report

Export XML



Export XML

For direct, machine-readable transmission to regulatory verifiers.



Print PDF

Consolidated, presentation-ready reports for internal management and C-suite review.

The unified ecosystem for absolute EEXI and CII mastery



NavCIIFuel transforms the burden of maritime compliance into a strategic, data-driven operational advantage.

A Unified Compliance Ecosystem



FuelEU: The Financial Frontier

Moving beyond measurement to financial strategy under strict EU regulatory frameworks.

FuelEU transforms carbon intensity directly into financial penalty or competitive advantage.
Precision is no longer optional.



Automated Voyage Classification Pipeline

Ingestion



Drag-and-drop Excel upload (handling up to 10,000 voyage rows).

[Download template](#)

Processing Engine



Built-in Port Classification Registry
(UN/LOCODE cross-referencing)



EEA to EEA
(100% MRV scope)
- 16 voyages.



**EEA to Non-EEA /
Non-EEA to EEA**
(50% MRV scope)
- 19 voyages total.



EEA Port Calls
(In-port emissions)
- 14 voyages.

The End-to-End Compliance Lifecycle

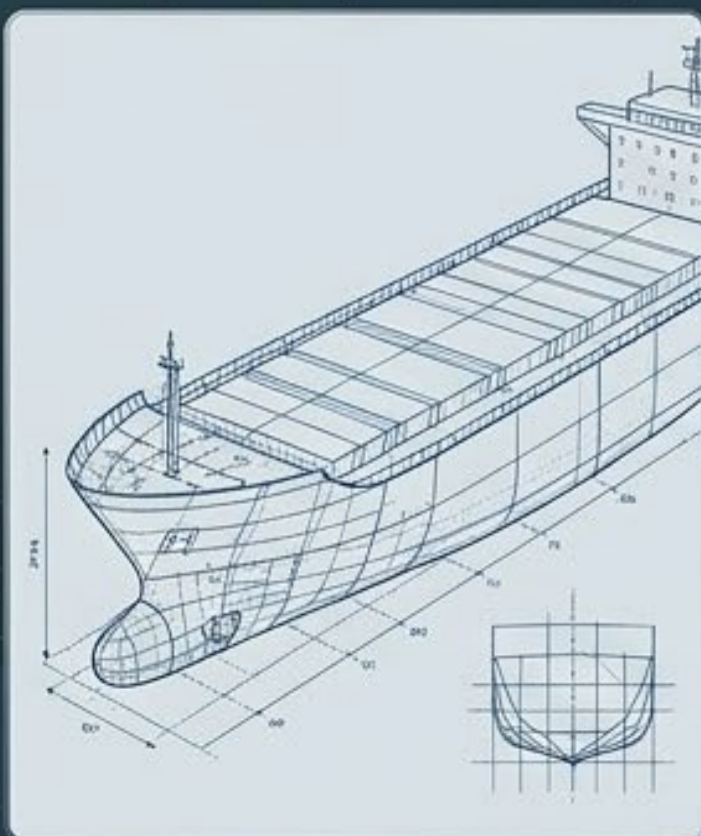


The NavCIIfuel Ecosystem

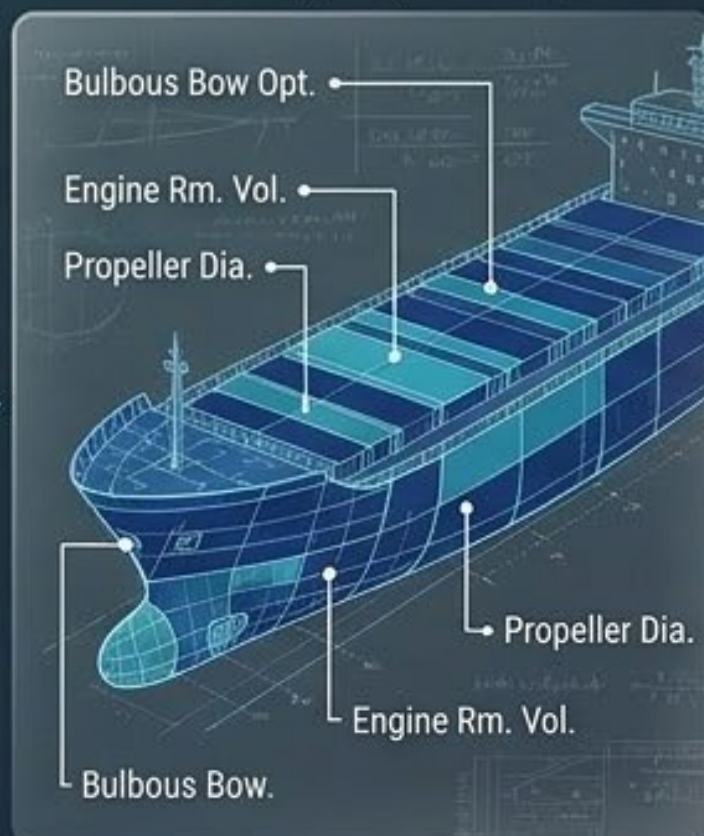
Total alignment. Every calculation and regulation housed under one intelligent roof.

Blueprint to Ocean Transition

Physics (NavStab)



Design (EEXI)



Operations (CII)



Strategy (FuelEU)



Data entered once propagates perfectly through every regulatory framework—saving time, eliminating manual data entry, and preventing costly compliance errors.

Access the definitive tool for maritime decarbonization



Get in touch for calculator support :
navciifuel@gmail.com / www.navciifuel.in